INCH-POUND
MIL-DTL-24227A(SH)
13 March 2018
SUPERSEDING
MIL-F-24227(SH)
18 May 1966

DETAIL SPECIFICATION

FITTINGS AND FLANGES, CAST BRONZE, SILVER-BRAZING, SUITABLE FOR ULTRASONIC INSPECTION

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tube fittings and flanges of cast bronze with ends for silver-brazing into a piping system suitable for ultrasonic inspection. Fittings are interior streamlined from socket-to-socket for use with inserted rings of silver-brazing alloy. Flanges are suitable for use with inserted rings of silver-brazing alloy. These fittings and flanges are not suitable for use in drinking (potable) water systems (see 3.1).

1.2 Classification. Fittings and flanges are of the following types, as specified (see 6.2).

Fittings and flanges		Figure	Table
Fittings	Caps	17	XX
	Couplings and reducers		
	Reducer (reducing one size)	7	X
	Reducer (reducing two sizes)	8	X
	Straight coupling	6	X
	Crosses		
	Straight	4	IX
	Elbows		
	90° straight	2	IX
	45° straight	5	IX
	Plugs-square		
	Cored	15	XIX
	Solid	16	XIX

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4730

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

Fittings and flanges		Figure	Table
Fittings (cont.)	Tees		
	Reducing	9	XI
	Straight	3	IX
	Unions		
	Nut	13	XVIII
	Tailpiece	14	XVIII
	Thread piece	12	XVIII
Flanges	150-pound series	10	XII, XV, XVI
	250-pound series	10	XIII, XV, XVI
	400-pound series	10	XIV, XV, XVI
Socket details		1	VIII
Silver-brazing ring		11	XVII

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-B-654 - Brazing Alloys, Silver

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-798 - Nondestructive Testing, Welding, Quality Control, Material Control and Identification and HI-Shock Test Requirements for Piping System Components for Naval Shipboard Use

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

NSF/ANSI 61 - Drinking Water System Components – Health Effects

(Copies of this document are available online at http://webstore.ansi.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

ASME Y14.5 - Dimensioning and Tolerancing

(Copies of these documents are available online at www.asme.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM B61	-	Standard Specification for Steam or Valve Bronze Castings
ASTM B271/B271M	-	Standard Specification for Copper-Base Alloy Centrifugal Castings
ASTM B505/B505M	-	Standard Specification for Copper Alloy Continuous Castings
ASTM B584	-	Standard Specification for Copper Alloy Sand Castings for General Applications

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Unless otherwise specified (see 6.2), fittings and flanges shall be made of material conforming to ASTM B61 (UNS C92200). Material conforming to alloy C90300 or C92200 of ASTM B271/B271M, ASTM B505/B505M, or ASTM B584 is an acceptable substitute for material conforming to ASTM B61 (alloy C92200). Fittings and flanges manufactured using alloys C90300 and C92200 do not meet the definition of "lead free" in accordance with NSF/ANSI 61 and, therefore, shall not be used for potable water systems.

3.1.1 Continuous melt. Castings produced by the continuous melt process and not identified with a heat number or code may be accepted provided the controls are approved by the procuring activity and cognizant inspector. The types of sampling and metallurgical controls described in ASTM B61 and ASTM B505/B505M used for continuous process controls are acceptable provided the following conditions of review and approval are met and maintained:

a. Quality control shall be in accordance with MIL-STD-798.

b. Material control shall be in accordance with MIL-STD-798.

3.2 Working pressure and temperature. Maximum allowable working pressures and temperatures shall be as specified in table I and table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE I. Maximum working pressure and temperature for fittings.

Fittings	Steam service		Water, oil, or gas service	
	Maximum working pressure (psi)	Maximum temperature (°F)	Maximum working pressure (psi)	Maximum temperature (°F)
All fittings above 6-inch size	150	425	250	150
Unions and all fittings up to and including 6-inch size	200	425	400	150

TABLE II. Maximum working pressure and temperature for flanges.

Series (pounds)	NPS (inches)	Rating (water, oil, gas, or steam)
150	½ to 12, inclusive	150 psi at 150 °F 100 psi at 425 °F
250	½ to 12, inclusive	250 psi at 150 °F 150 psi at 425 °F
400	½ to 10, inclusive	400 psi at 150 °F 200 psi at 425 °F

3.3 Design (fittings and flanges).

3.3.1 Surface finish.

3.3.1.1 Machined surfaces. Lugs, ribs, or flats may be cast on fittings for holding during machining provided placement does not interfere with ultrasonic inspection. All machined surfaces marked with "✓" shall be 125 microinches (µin) (3.2 µm) roughness average (Ra) or smoother in accordance with ASME B46.1. The surface finish of the union seat shall be 63 µin (1.6 µm) Ra or smoother. After machining, all fittings and flanges shall be cleaned to remove all foreign particles. All sharp edges, except for edges at the face of the socket, shall be removed.

3.3.1.2 As cast surfaces. External surfaces may be supplied as cast, provided they are cast true and smooth. Fins that are not well-rounded, all sharp edges (except for edges at the face of the socket), and all burrs shall be blended.

3.3.2 Silver-brazing rings. Pre-inserted rings shall be in accordance with the requirements for BCuP-5 of QQ-B-654. Dimensions shall be as specified in 3.3.4.2. Fittings and flanges shall be supplied without pre-inserted silver-brazing rings inserted.

3.3.3 Measurements. Dimensioning, tolerancing, and control symbols shall be in accordance with ASME Y14.5.

3.3.4 Dimensions.

3.3.4.1 Dimensions for fittings and flanges. Fitting and flange dimensions shall be in accordance with figures 1 through 17 and tables III through XX, as applicable. Within any one fitting, the axis of each silver-braze socket shall not deviate from true position with the axis of any other silver-braze socket by more than 1/16 inch per foot.

3.3.4.2 Dimensions for pre-inserted rings. Dimensions for pre-inserted rings shall be as shown on figure 11.

3.3.5 Tolerances.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

3.3.5.1 Tolerances for fittings. The tolerances specified in table III shall be permitted in center-to-end dimensions for all fittings.

TABLE III. Tolerances for fittings, inches.

NPS	Tolerance ±	NPS	Tolerance ±
½	0.06	3	0.10
¾	0.06	3½	0.10
1	0.07	4	0.12
1¼	0.07	5	0.12
1½	0.08	6	0.12
2	0.08	8	0.15
2½	0.10	10	0.20
---	---	12	0.20

3.3.5.2 Tolerances for flanges. Tolerances for flanges shall be as shown in table IV.

TABLE IV. Tolerances for flanges, inches (see figure 10).

NPS	W	T	S	O
1¼ and under	±1/16	±0.010	+0.003 -0.000	+0.000 -0.031
1½ to 2	±1/16	±0.010	+0.005 -0.000	+0.000 -0.031
2½ to 12	±1/16	±0.010	+0.007 -0.000	+0.000 -0.047

3.3.5.3 Tolerances for general dimensions. Unless otherwise specified (see 6.2), tolerances shall be as specified in table V.

TABLE V. Tolerances for general dimensions, inches.

Dimensional tolerances							Casting tolerances	
Machined				Geometric (where indicated)				
Dimensions	Decimal		Fraction	Sym	Geometry	Tolerances	Dimensions	Tolerances
	2 PL	3 PL						
				⌒	Flat and Straight	0.0005	Under 2	±1/16
Under 6	±0.010	±0.005	±1/64	⊥	Perpendicularity	0.001	2 to 5	+3/32, -1/16
6 to 24	±0.020	±0.010	±1/32	//	Parallelism	0.001	5 to 8	+1/8, -3/32
24 and over	±0.030	±0.015	±1/16	◎	Concentricity	As noted	8 to 12	+3/16, -1/8
Fillets and radii - 1/8 inch				---			12 to 15	+1/4, -3/16
Tolerances on angles to be ±0° - 30 inches				Control symbol – ASME Y14.5			15 and over	±1/4

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

3.3.6 Reducing fittings. The wall thickness of reducing fittings shall be that required for the sizes specified (see 6.2 and table VIII).

3.3.7 45-degree angles. Except in the crotch area of elbows, tees, and crosses, the 45-degree angle on the fitting socket outside diameter (OD) may be removed by continuing dimension "S" off the back of the socket hub (see figure 1).

3.3.8 Sockets. All fittings shall have sockets in accordance with table VIII.

3.3.9 Threaded pieces. Threaded pieces for unions shall have octagonal or round sections knurled or ribbed for the use of wrenches (see figure 12).

3.3.10 Brazing ring grooves. The thinnest flange socket wall caused by machining the pre-inserted ring groove shall be not less than ¼ the difference between dimensions "E" and "S" as shown on figure 10.

3.3.11 Flanges.

3.3.11.1 Flange finish. Flange faces shall have a finish with a circular lay (concentric or phonographic) of the 125 µin (3.2 µm) to 1,000 µin (25 µm) Ra produced by machining 30 to 80 cuts of uniform depth per inch of face width. The bore of the hub (diameter "S" of figure 10) shall have an Ra of 125 µin (3.2 µm) or better. Unless otherwise specified herein, the flanges shall have a waterway (diameter "T" of figure 10) machined with an Ra finish of 125 µin (3.2 µm) to the bore specified (see 6.2). Flanges shall not be finished on the rim provided that they are cast smooth and true. The remaining external surfaces shall not be finished except for ultrasonic inspection area and spot faces.

3.3.11.2 Flange drilling. Unless otherwise specified (see 6.2), flanges shall be undrilled. Drilled flanges shall be spot-faced on the back parallel to the face of the flange for the bolt-heads and nuts. The spot-face shall be as specified in table VI. The spot face shall cut not more than 1/32 inch below the minimum flange thickness specified in tables XII, XIII, and XIV.

TABLE VI. Spot-face dimensions, inches.

Bolt hole diameter	Spot-face diameter
9/16	1 1/16
11/16	1¼
13/16	1 7/16
15/16	1 13/16
1 1/16	2 1/16

3.4 Identification. Each fitting, union, and flange shall be identified by casting, electro-chemical etch, vibro tool etch, or stamping with the following:

a. Manufacturer's name or trademark
b. Nominal pipe size (NPS)
c. Rating:
   (1) Fittings:
      (a) ½ inch through 6 inches – 400 WOG
      (b) 6 inches through 12 inches – 250 WOG
   (2) Flanges (as applicable):
      (a) 150 WOG

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

(b) 250 WOG

(c) 400 WOG

d. BRZ for bronze

e. Heat number or code (see 3.1.1)

3.4.1 Stamping. All stamping shall be done with round bottom stamps.

3.4.2 Location. Markings on the fittings and flanges shall be applied so as not to reduce the wall thickness below the minimum and so as not to affect the strength of the fitting. Placement shall not interfere with ultrasonic inspection.

3.4.2.1 Flange marking location. Identification markings shall be provided on the rim of all flanges. Cast identification marking is allowable on the flange back, provided placement does not subject marking to removal during machining.

3.5 Workmanship. All fittings and flanges shall be sound, smoothly cored, true to form, uniform in texture, and free from cold shuts, porosity, or any other defects which may affect their serviceability. They shall be thoroughly cleaned, both inside and outside.

4. VERIFICATION

4.1 Lot. All fittings or flanges of the same size offered for delivery at one time shall be considered a lot for purposes of conformance inspection.

4.2 Sampling. A random sampling of fittings or flanges shall be selected from each lot for the examination specified in 4.3 and the test specified in 4.4.2 in accordance with ANSI/ASQ Z1.4 at inspection level II and the acceptance quality limit (AQL), as specified (see 6.2).

4.3 Examination. Each of the samples selected in accordance with 4.2 shall be examined to verify compliance with all the requirements of this specification not involving tests. Examination shall be conducted as specified in table VII. Any fittings or flanges in the sample containing one or more defects shall be rejected, and if the number of defective fittings or flanges in any sample exceeds the AQL specified (see 6.2), the lot represented by that sample shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE VII. Classification of defects.

Categories	Defects
Critical Defects	
	None defined.
Major Defects	
101	Ultrasonic inspection bands damaged or not as specified.
102	Fitting/Flange, type and size of fitting and series and size of flange not as specified.
103	Material not as specified.
104	Fitting or flange not sound, smoothly cored, true to form, uniform in texture, not free from cold shuts and porosity.
105	Fitting or flange surface (internal and external) not thoroughly cleaned, fins and burrs not rounded or blended.
106	Unauthorized repairs, plugs, welds, or impregnation.
107	Thread missing, torn, stripped, broken, or not complete.
108	Thread size not as specified.
109	Socket not smooth.
110	Socket dimensions not as specified.
111	Bottom of socket not streamlined.
112	Bottom of socket not square with axis of fitting.
113	Dimensions not within the tolerances specified.
114	Threaded pieces of unions not octagonal, knurled, or ribbed.
115	Reducing fitting wall thickness not as specified.
Minor Defects	
201	Marking, manufacturer's name or trademark missing, not cast, incorrect, or illegible.

4.4 Conformance inspection.

4.4.1 Air pressure leakage test. Each fitting or flange shall be tested underwater or with soapsuds at 100 pounds per square inch (psi) air for 10 seconds. No leakage is allowed.

4.4.2 Hydrostatic pressure leakage test. Each sample fitting or flange selected in accordance with 4.2 shall be subjected to a hydrostatic pressure 1½ times the working pressure for 1 minute. Under hydrostatic pressure, the fitting or flange shall not leak water or sweat at any part of the surface.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Fittings and flanges herein are intended for silver-brazing installed and ultrasonically inspected on steam, gas, oil, or water piping services aboard naval vessels. These fittings and flanges are intended for general service and specifically for seawater service where ultrasonic inspection for joint integrity is required. Existing stock of fittings and flanges manufactured to MIL-F-24227 dated 18 May 1966 are acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Quantity required.
c. Type and size of fitting required and series and size of flange required (see 1.2).
d. Material requirements if other than as specified (see 3.1).
e. Tolerances for general dimensions if other than as specified (see 3.3.5.3).
f. Reducing fitting requirements (see 3.3.6).
g. Flange waterway bore required (3.3.11.1).
h. When flanges are to be drilled (see 3.3.11.2).
i. Requirement for an AQL of 1.0 (see 4.2).
j. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Caps
Cross
Elbow
Plug
Tee
Union

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

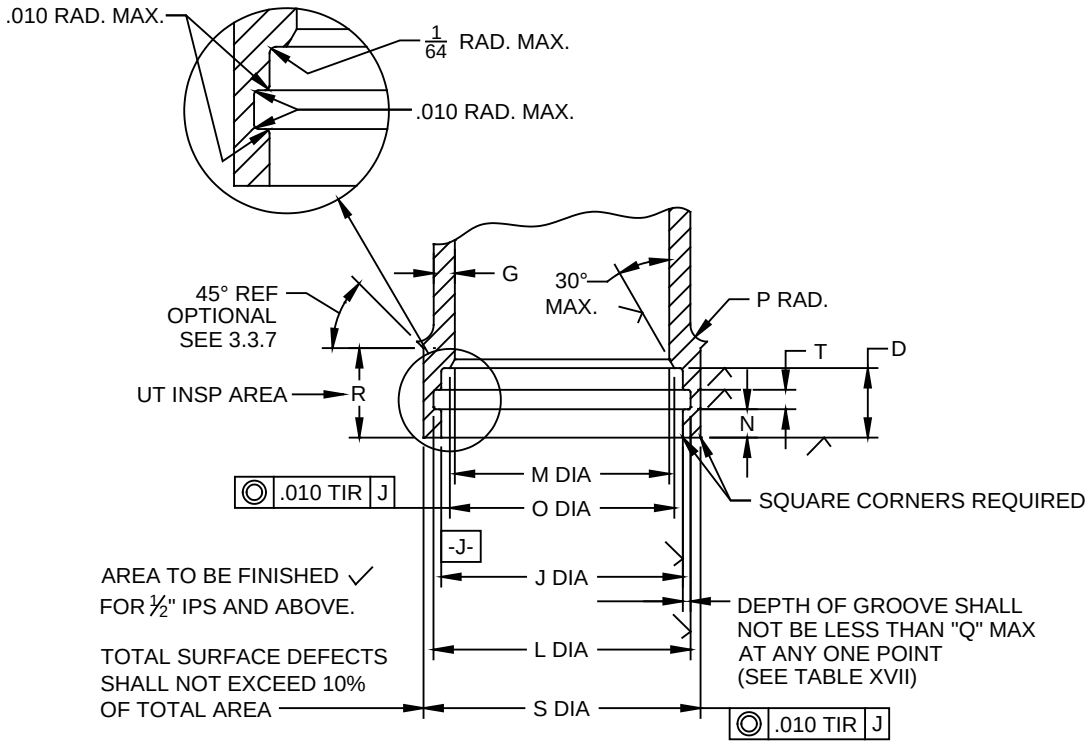


FIGURE 1. Socket detail (see table VIII).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE VIII. Detail dimensions of socket (see figure 1), inches.

NPS	OD of tube (Max.)	D	R	M		G	P	S	J		L	T	N	O
		Depth of socket +1/64 -0	Width of UT band +1/16 -0	Inside diameter of fittings		Metal thickness	Radius (Min.)	Band of diameter +1/8 -0 (Min.)	Diameter of socket		Diameter of groove +0.020 -0.000	Width of groove +0.010 -0.000	Fitting face of ring groove	Chamfered bore ±0.010
				---	±				Min.	Max.				
¼	0.540	17/64	---	0.398	0.015	0.070	5/64	0.700	0.540	0.543	0.619	0.068	9/64	---
⅜	0.675	5/16	---	0.532	0.015	0.080	3/32	0.855	0.875	0.678	0.766	0.083	9/64	---
½	0.840	⅜	33/64	0.697	0.020	0.080	3/32	1.020	0.840	0.843	0.929	0.083	9/64	0.707
¾	1.050	13/32	36/64	0.907	0.025	0.090	7/64	1.250	1.050	1.053	1.159	0.099	5/32	0.962
1	1.315	7/16	19/32	1.171	0.025	0.100	7/64	1.535	1.318	1.318	1.424	0.130	5/32	1.166
1¼	1.660	½	41/64	1.502	0.030	0.110	⅛	1.900	1.660	1.663	1.789	0.130	3/16	1.522
1½	1.900	⅝	49/64	1.742	0.030	0.120	⅛	2.160	1.900	1.905	2.011	0.192	7/32	1.762
2	2.375	21/32	51/64	2.186	0.030	0.140	5/32	2.675	2.375	2.380	2.488	0.192	15/64	2.206
2½	2.875	25/32	59/64	2.688	0.030	0.150	5/32	3.215	2.875	2.882	2.988	0.192	19/64	2.706
3	3.500	53/64	31/32	3.286	0.035	0.170	3/16	3.880	3.500	3.507	3.615	0.255	19/64	3.311
3½	4.000	⅞	1 1/64	3.786	0.035	0.180	3/16	4.400	4.000	4.007	4.118	0.255	5/16	3.811
4	4.500	29/32	1 1/16	4.252	0.040	0.200	7/32	4.940	4.500	4.507	4.678	0.255	31/64	4.282
5	5.563	1	1 9/64	5.278	0.040	0.220	5/16	6.163	5.563	5.570	5.736	0.255	⅜	5.308
6	6.625	1 7/64	1¼	6.321	0.040	0.320	11/32	7.305	8.625	8.632	8.788	0.255	27/64	6.351
8	8.625	1 5/16	1 29/64	8.286	0.040	0.330	13/32	8.625	8.625	8.632	8.788	0.398	29/64	8.316
10	10.750	1½	1 41/64	10.325	0.040	0.445	½	11.700	10.750	10.757	10.933	0.398	35/64	10.355
12	12.750	1⅝	1 49/64	12.322	0.040	0.540	9/16	13.840	12.750	12.757	12.963	0.398	39/64	12.352

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

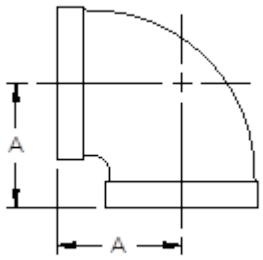


FIGURE 2. Elbow straight, 90° (see table IX).

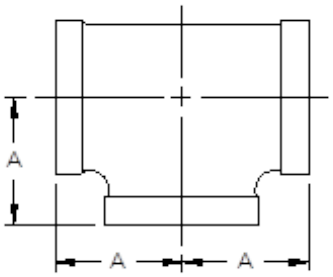


FIGURE 3. Tee, straight (see table IX).

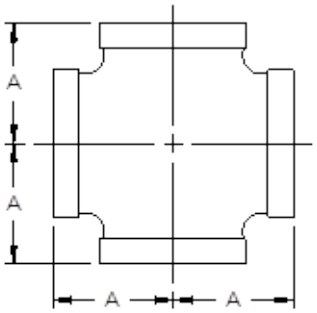


FIGURE 4. Cross, straight (see table IX).

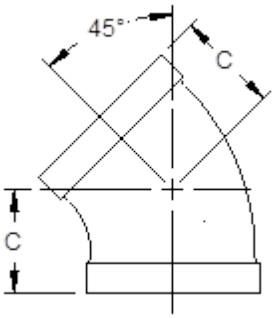


FIGURE 5. Elbow straight, 45° (see table IX).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE IX. Center-to-end dimensions of elbows straight 90°, tees straight, crosses straight, elbows straight 45°, inches (see table VIII for detail dimensions).

NPS	A	C
	Center-to-end 90° elbows, tees, and crosses	Center-to-end 45° elbows
½	1.244	1.014
¾	1.414	1.124
1	1.649	1.279
1¼	1.909	1.439
1½	2.153	1.519
2	2.526	1.668
2½	2.919	2.169
3	3.299	2.389
3½	3.639	2.609
4	4.010	2.830
5	4.719	3.269
6	5.442	3.678
8	6.763	4.483
10	8.268	5.348
12	9.687	6.157

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

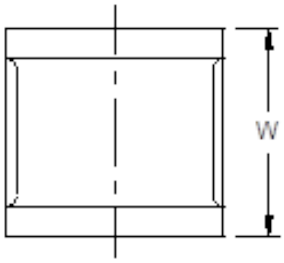


FIGURE 6. Straight coupling (see table X).

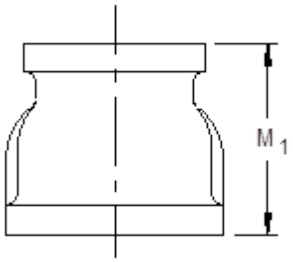


FIGURE 7. Reducer (reducing one size) (see table X).

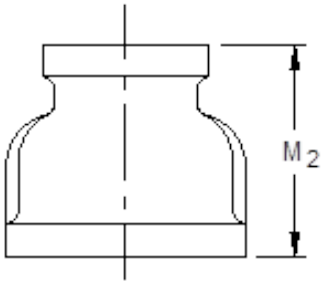


FIGURE 8. Reducer (reducing two sizes) (see table X).

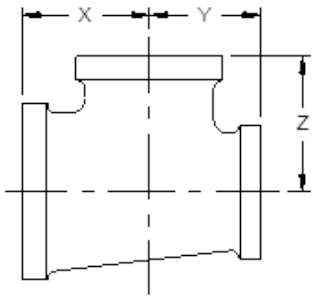


FIGURE 9. Tee, reducing (see table XI).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE X. End-to-end dimensions of cast couplings and reducers, inches (see table VIII for detail dimensions).

NPS	W End-to-end (straight)	M1 End-to-end (reducing one size)	M2 End-to-end (reducing two sizes)	Size of reduced ends
¼	1.252	---	---	---
⅜	1.330	1.134	---	¼
½	1.570	1.450	1.254	¼, ⅜
¾	1.710	1.640	1.520	⅜, ½
1	1.930	1.825	1.755	½, ¾
1¼	2.110	2.013	1.915	¾, 1
1½	2.170	2.140	2.050	1, 1¼
2	2.448	2.309	2.279	1¼, 1½
2½	3.130	3.499	3.530	1½, 2
3	3.430	3.940	3.940	2, 2½
3½	3.680	4.250	4.250	2½, 3
4	3.942	4.631	4.631	3, 3½
5	4.470	5.231	5.243	3½, 4
6	4.998	---	---	---
8	5.968	---	---	---
10	6.688	---	---	---
12	7.186	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE XI. Center-to-end dimensions of reducing tees, inches (see table VIII for detail dimensions) (for tolerances, see table III).

NPS	Center-to-end		
	X	Y	Z
½ by ½ by ¼	1.053	1.053	1.134
½ by ¼ by ½	1.244	1.134	1.244
¾ by ¾ by ½	1.307	1.307	1.359
¾ by ¾ by ⅜	1.194	1.194	1.296
¾ by ¾ by ¼	1.116	1.116	1.249
¾ by ½ by ¾	1.422	1.359	1.422
¾ by ½ by ½	1.307	1.244	1.359
¾ by ⅜ by ¾	1.422	1.296	1.422
¾ by ⅜ by ⅜	1.194	1.068	1.296
⅜ by ⅜ by ½	1.181	1.181	1.131
½ by ⅜ by ⅜	1.131	1.068	1.181
½ by ½ by ⅜	1.131	1.131	1.181
½ by ½ by ¾	1.359	1.359	1.307
¾ by ⅜ by ½	1.307	1.181	1.359
¾ by ¾ by 1	1.565	1.565	1.500
1 by ⅜ by 1	1.643	1.439	1.643
1 by ½ by ¾	1.500	1.359	1.565
1 by ¾ by ⅜	1.272	1.194	1.439
1 by 1 by ⅜	1.272	1.272	1.439
1 by 1 by ¾	1.500	1.500	1.565
1 by 1 by ½	1.385	1.385	1.502
1 by 1 by ¼	1.194	1.194	1.392
1 by 1 by 1¼	1.825	1.825	1.721
1 by ¾ by 1	1.643	1.565	1.843
1 by ¾ by ¾	1.500	1.422	1.565
1 by ¾ by ½	1.385	1.307	1.502
1 by ½ by 1	1.643	1.502	1.643
1 by ½ by ½	1.385	1.244	1.502
1¼ by ½ by 1	1.721	1.534	1.825
1¼ by 1¼ by ⅜	1.350	1.350	1.621
1¼ by 1¼ by 1	1.721	1.721	1.825
1¼ by 1¼ by ¾	1.578	1.578	1.747
1¼ by 1¼ by ½	1.463	1.463	1.716

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE XI. Center-to-end dimensions of reducing tees, inches (see table VIII for detail dimensions) (for tolerances, see table III) – Continued.

NPS	Center-to-end		
	X	Y	Z
1¼ by 1 by 1¼	1.903	1.825	1.903
1¼ by 1 by 1	1.721	1.643	1.825
1¼ by 1 by ¾	1.578	1.500	1.747
1¼ by ¾ by 1¼	1.903	1.747	1.903
1¼ by ¾ by 1	1.721	1.565	1.825
1¼ by ¾ by ¾	1.578	1.422	1.747
1½ by ¾ by ¾	1.703	1.422	1.877
1½ by 1 by 1	1.846	1.643	1.955
1½ by 1 by 1¼	2.028	1.825	2.033
1½ by 1¼ by ½	1.588	1.463	1.846
1½ by 1½ by 1¼	2.028	2.028	2.033
1½ by 1½ by 1	1.846	1.846	1.955
1½ by 1½ by ¾	1.703	1.703	1.877
1½ by 1½ by ½	1.588	1.588	1.846
1½ by 1½ by ⅜	1.475	1.475	1.751
1½ by 1¼ by 1½	2.158	2.033	2.158
1½ by 1¼ by 1¼	2.028	1.903	2.033
1½ by 1¼ by 1	1.846	1.721	1.955
1½ by 1 by 1½	2.158	1.955	2.158
1½ by ¾ by 1½	2.158	1.908	2.158
1½ by ¾ by 1¼	2.028	1.747	2.033
1½ by ¾ by 1	1.846	1.565	1.955
1½ by 1½ by 1½	2.158	1.846	2.158
2 by ¾ by 2	2.510	2.166	2.510
2 by 1½ by 1¼	2.122	2.028	2.291
2 by 2 by 2½	2.252	2.252	2.416
2 by 2 by 1¼	2.122	2.122	2.291
2 by 2 by 1	1.940	1.940	2.244
2 by 1½ by 1½	2.252	2.158	2.416

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE XI. Center-to-end dimensions of reducing tees, inches (see table VIII for detail dimensions) (for tolerances, see table III) – Continued.

NPS	Center-to-end		
	X	Y	Z
2 by 1½ by 1	1.940	1.846	2.244
2 by 1½ by 2	2.510	2.416	2.510
2 by 1¼ by 2	2.510	2.291	2.510
2 by 1¼ by 1½	2.252	2.033	2.416
2 by 1¼ by 1¼	2.122	1.903	2.291
2 by 1 by 2	2.510	2.213	2.510
2 by ½ by 2	2.510	2.135	2.510
2½ by 2½ by 2	2.635	2.635	2.780
2½ by 2½ by 1	2.065	2.065	2.483
2½ by 2½ by ¾	1.922	1.922	2.436
2½ by 2 by 1½	2.377	2.252	2.686
2½ by 1½ by 2	2.635	2.416	2.780
2½ by 1½ by 1½	2.377	2.158	2.686
2½ by ½ by 2½	2.905	2.405	2.905
2½ by 2 by 2	2.636	2.510	2.780
2½ by 2½ by 1½	2.377	2.377	2.686
3 by 3 by 2½	2.952	2.952	3.237
3 by 3 by 2	2.682	2.682	3.112
3 by 3 by 1½	2.424	2.424	3.018
3 by 2½ by 2½	2.952	2.905	3.237
3 by 2½ by 2	2.682	2.635	3.112
3 by 2 by 3	3.284	3.112	3.284
3 by 2 by 2	2.684	2.510	3.112
3½ by 3½ by 3	3.362	3.362	3.544
3½ by 3½ by 2½	3.030	3.030	3.497
3½ by 3 by 2½	3.030	2.952	3.497
3½ by 3½ by 2½	3.030	2.905	3.497
4 by 4 by 3½	3.731	3.731	3.892
4 by 4 by 3	3.471	3.471	3.814
4 by 3 by 3	3.471	3.284	3.814
5 by 5 by 4	4.111	4.111	4.613
5 by 3 by 4	4.111	3.814	4.613
6 by 6 by 5	4.863	4.863	5.294

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE XI. Center-to-end dimensions of reducing tees, inches (see table VIII for detail dimensions) (for tolerances, see table III) – Continued.

NPS	Center-to-end		
	X	Y	Z
6 by 6 by 4	4.251	4.251	5.814
8 by 8 by 6	5.699	5.699	6.494
10 by 10 by 8	7.135	7.135	7.896
12 by 12 by 8	7.479	7.479	8.966

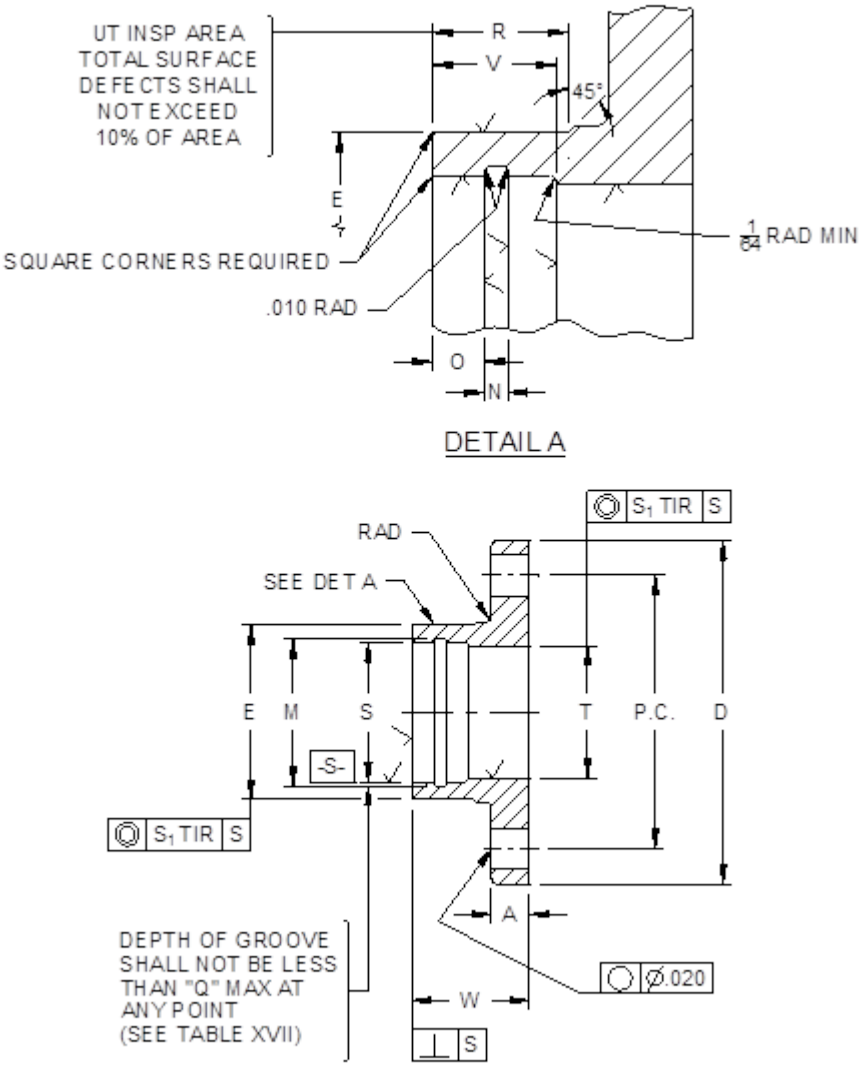


FIGURE 10. Detail dimensions of flanges (see table XII, table XIII, table XIV, table XV, and table XVI).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE XII. Dimensions of 150-pound flanges, inches (see figure 10).

NPS	Tubing OD	A +1/8 -1/64	R +1/16 -0	D	E		S	T	V +1/64 -0	W	RAD	S1
					Diameter	Tol						
1/2	0.840	3/8	0.500	3 9/16	1 5/32	+1/8 -0	0.840	0.71	3/8	1.219	1/4	0.008
3/4	1.050	7/16	0.531	3 13/16	1 11/32	+1/8 -0	1.050	0.92	13/32	1.313	1/4	0.008
1	1.315	7/16	0.563	4 1/4	1 5/8	+1/8 -0	1.315	1.18	7/16	1.345	1/4	0.016
1 1/4	1.660	7/16	0.625	4 1/2	2	+1/8 -0	1.660	1.53	1/2	1.469	5/16	0.016
1 1/2	1.900	7/16	0.750	5 1/16	2 1/4	+1/8 -1	1.900	1.77	5/8	1.594	5/16	0.016
2	2.375	7/16	0.781	5 9/16	2 3/4	+1/8 -0	2.375	2.24	21/32	1.625	5/16	0.016
2 1/2	2.875	1/2	0.750	6 1/8	3 1/4	+1/8 -0	2.875	2.71	5/8	1.656	5/16	0.031
3	3.500	1/2	0.750	6 5/8	3 7/8	+1/8 -0	3.500	3.31	5/8	1.656	5/16	0.031
3 1/2	4.000	1/2	0.813	7 3/16	4 3/8	+3/16 -0	4.000	3.81	11/16	1.782	3/8	0.031
4	4.500	1/2	0.813	7 11/16	4 5/16	+3/16 -0	4.500	4.28	11/16	1.782	3/8	0.031
5	5.563	9/16	0.813	9 1/16	6	+3/16 -0	5.563	5.32	11/16	1.844	3/8	0.047
6	6.625	9/16	0.875	10 1/8	7 1/16	+3/16 -0	6.625	6.36	3/4	1.969	7/16	0.047
8	8.625	5/8	1.063	12 3/8	9 1/8	+3/16 -0	8.625	8.33	15/16	2.220	7/16	0.063
10	10.750	11/16	1.187	15	11 5/16	+3/16 -0	10.750	10.38	1 1/16	2.419	1/2	0.063
12	12.750	3/4	1.375	17 5/8	13 7/16	+3/16 -0	12.750	12.25	1 1/4	2.719	1/2	0.063

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE XIII. Dimensions of 250-pound flanges, inches (see figure 10).

NPS	Tubing OD	A +1/8 -1/64	R +1/16 -0	D	E		S	T	V +1/64 -0	W	RAD	S1
					Diameter	Tol.						
1/2	0.840	11/16	0.500	3 9/16	1 5/32	+1/8 -0	0.840	0.697	3/8	1.532	1/4	0.008
3/4	1.050	11/16	0.531	3 9/16	1 11/32	+1/8 -0	1.050	0.907	13/32	1.563	1/4	0.008
1	1.315	3/4	0.563	4 1/4	1 5/8	+1/8 -0	1.315	1.171	7/16	1.657	1/4	0.016
1 1/4	1.660	13/16	0.625	4 1/2	2	+1/8 -0	1.660	1.502	1/2	1.843	5/16	0.016
1 1/2	1.900	13/16	0.750	5 1/16	2 1/4	+1/8 -0	1.900	1.742	5/8	1.968	5/16	0.016
2	2.375	13/16	0.781	5 9/16	2 3/4	+1/8 -0	2.375	2.186	21/32	1.999	5/16	0.016
2 1/2	2.875	15/16	0.750	6 1/8	3 1/4	+1/8 -0	2.875	2.686	5/8	2.094	5/16	0.031
3	3.500	15/16	0.750	6 5/8	3 7/8	+1/8 -0	3.500	3.286	5/8	2.094	5/16	0.031
3 1/2	4.000	1	0.813	7 3/16	4 3/8	+3/16 -0	4.000	3.786	11/16	2.282	3/8	0.031
4	4.500	1	0.813	7 11/16	4 15/16	+3/16 -0	4.500	4.252	11/16	2.282	3/8	0.031
5	5.563	1 1/16	0.813	9 1/16	6	+3/16 -0	5.563	5.276	11/16	2.344	3/8	0.047
6	6.625	1 3/16	0.875	10 1/8	7 1/16	+3/16 -0	6.625	6.321	3/4	2.595	7/16	0.047
8	8.625	1 5/16	1.063	12 3/8	9 1/8	+3/16 -0	8.625	8.286	15/16	2.907	7/16	0.063
10	10.750	1 7/16	1.187	15	11 5/16	+3/16 -0	10.750	10.325	1 1/16	3.219	1/2	0.063
12	12.750	1 1/2	1.375	17 5/8	13 7/16	+3/16 -0	12.750	12.322	1 1/4	3.569	1/2	0.063

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE XIV. Dimensions of 400-pound flanges, inches (see figure 10).

NPS	Tubing OD	A +⅛ -1/64	R +1/16 -0	D	E		S	T	V +1/64 -0	W	RAD	S1
					Diameter	Tol.						
½	0.840	11/16	0.500	4	1¼	+⅛ -0	0.840	0.71	⅜	1.532	¼	0.008
¾	1.050	11/16	0.531	4 5/16	1 7/16	+⅛ -0	1.050	0.92	13/32	1.563	¼	0.008
1	1.315	¾	0.563	5 1/16	1¾	+⅛ -0	1.315	1.18	7/16	1.657	¼	0.016
1¼	1.660	13/16	0.625	5⅜	2⅛	+⅛ -0	1.660	1.53	½	1.843	15/16	0.016
1½	1.900	13/16	0.750	5 15/16	2⅜	+⅛ -0	1.900	1.77	⅝	1.962	15/16	0.016
2	2.375	13/16	0.781	6½	2⅞	+⅛ -0	2.375	2.24	21/32	1.999	15/16	0.016
2½	2.875	15/16	1.125	7 9/16	3⅜	+⅛ -0	2.875	1	1	2.657	½	0.031
3	3.500	15/16	1.125	8⅛	4	+3/16 -0	3.500	1	1	2.657	½	0.031
3½	4.000	1	1.125	8 11/16	4½	+3/16 -0	4.000	1	1	2.719	½	0.031
4	4.500	1	1.250	9¼	5	+3/16 -0	4.500	1	1⅛	2.906	9/16	0.031
5	5.563	1 1/16	1.375	10⅜	6 3/16	+3/16 -0	5.563	1	1¼	3.093	9/16	0.047
6	6.625	1 3/16	1.500	11 15/16	7 5/16	+3/16 -0	6.625	1	1⅜	3.407	⅝	0.047
8	8.625	1 15/16	1.625	14¾	9⅜	+3/16 -0	8.625	1	1½	3.703	11/16	0.063
10	10.750	1 7/16	1.675	17	11⅝	+3/16 -0	10.75	1	1¾	4.157	¾	0.063

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE XV. Machining details for flange ring groove, inches (see figure 10).

NPS	Tubing OD	O		N	M
		150 and 250 pounds	400 pounds	+0.010 -0.000	+0.020 -0.000
½	0.840	0.149	0.149	0.083	0.929
¾	1.050	0.154	0.154	0.099	1.159
1	1.315	0.154	0.154	0.130	1.424
1¼	1.660	0.185	0.185	0.130	1.769
1½	1.900	0.217	0.217	0.192	2.011
2	2.375	0.232	0.232	0.192	2.486
2½	2.875	0.217	0.404	0.192	2.988
3	3.500	0.185	0.373	0.255	3.613
3½	4.000	0.217	0.373	0.255	4.113
4	4.500	0.217	0.435	0.255	4.673
5	5.563	0.217	0.498	0.255	5.736
6	6.625	0.248	0.560	0.255	6.798
8	8.625	0.270	0.551	0.398	8.798
10	10.750	0.333	0.676	0.398	10.923
12	12.750	0.426	---	0.398	12.963

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE XVI. Flange drilling dimensions, inches (see figure 10).

NPS	Tubing OD	Number of holes		150-pound flanges			250-pound flanges		400-pound flanges		
		150 and 250 pounds	400 pounds	Diameter of holes	Pitch circle	Pitch cord	Diameter of holes	Pitch circle	Diameter of holes	Pitch circle	Pitch cord
½	0.840	3	4	9/16	2 7/16	2.11	9/16	2 7/16	9/16	2 7/8	2.03
¾	1.050	4	4	9/16	2 11/16	1.90	9/16	2 11/16	9/16	3 3/16	2.25
1	1.315	4	5	9/16	3 1/8	2.21	9/16	3 1/8	11/16	3¾	2.20
1¼	1.660	4	5	9/16	3 3/8	2.39	9/16	3 3/8	11/16	4 1/16	2.39
1½	1.900	6	6	9/16	3 15/16	1.97	9/16	3 15/16	11/16	4 5/8	2.31
2	2.375	6	7	9/16	4 7/16	2.22	11/16	4 7/16	11/16	5 3/16	2.25
2½	2.875	6	8	9/16	5	2.50	11/16	5	13/16	6	2.30
3	3.500	8	8	9/16	5½	2.10	11/16	5½	13/16	6 9/16	2.51
3½	4.000	8	9	9/16	6 1/16	2.32	11/16	6 1/16	13/16	7 1/8	2.44
4	4.500	8	9	9/16	6 9/16	2.51	11/16	6 9/16	13/16	7 11/16	2.63
5	5.563	10	11	11/16	7 13/16	2.41	11/16	7 13/16	13/16	8 13/16	2.48
6	6.625	12	12	11/16	8 7/8	2.30	11/16	8 7/8	15/16	10 3/16	2.64
8	8.625	13	13	11/16	11 1/16	2.46	11/16	11 1/16	1 1/16	12¾	3.05
10	10.750	15	15	13/16	13 7/16	2.79	13/16	13 7/16	1 1/16	15	3.12
12	12.750	18	---	13/16	16 1/16	2.79	13/16	16 1/16	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

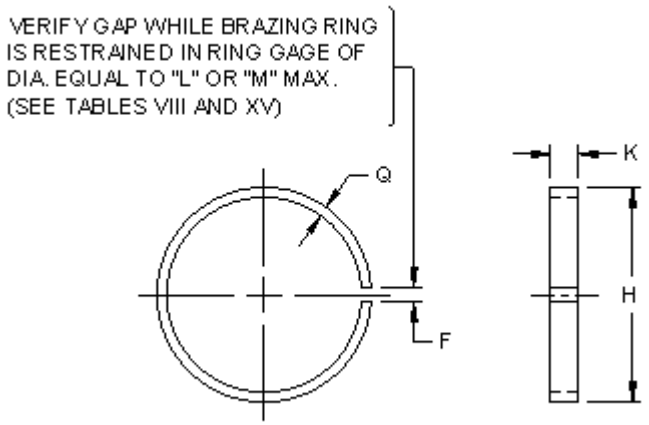


FIGURE 11. Silver-brazing ring.

TABLE XVII. Dimensions of silver-brazing ring, inches.

NPS	Q ±0.003	K ±0.003	H	F	
				Min.	Max.
½	0.040	0.078	0.949	0.073	0.130
¾	0.050	0.094	1.179	0.073	0.130
1	0.050	0.125	1.444	0.073	0.135
1¼	0.050	0.125	1.789	0.073	0.135
1½	0.050	0.188	2.031	0.073	0.135
2	0.050	0.188	2.506	0.073	0.135
2½	0.050	0.188	3.000	0.073	0.135
3	0.050	0.250	3.633	0.073	0.167
3½	0.050	0.250	4.133	0.073	0.167
4	0.080	0.250	4.693	0.073	0.229
5	0.080	0.250	5.756	0.073	0.229
6	0.080	0.250	6.818	0.073	0.261
8	0.080	0.391	8.818	0.073	0.261
10	0.080	0.391	10.943	0.073	0.261
12	0.100	0.391	12.983	0.073	0.261

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

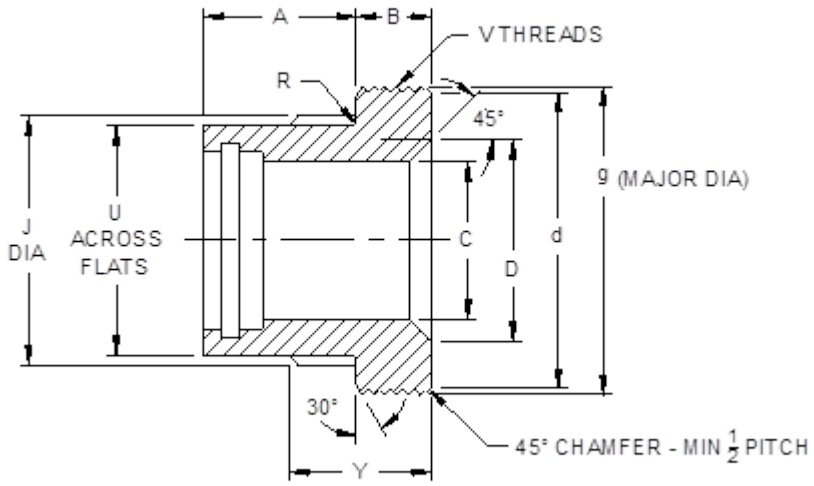


FIGURE 12. Thread piece (see table XVIII).

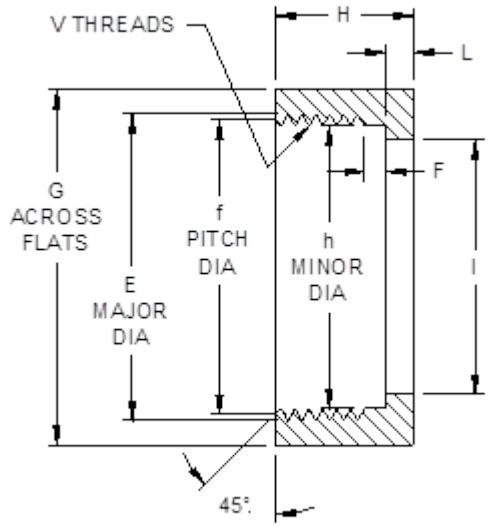


FIGURE 13. Union (see table XVIII).

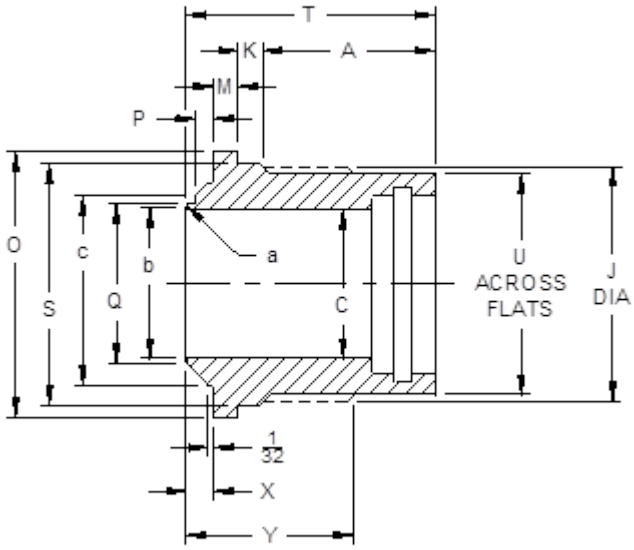


FIGURE 14. Tailpiece (see table XVIII).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Detail dimensions of unions, inches (see figure 12, figure 13, and figure 14).

Dimensions		NPS					
		½	¾	1	1¼	1½	2
A		1.006	1.085	1.197	1.379	1.594	1.775
B		25/64	½	½	35/64	35/64	35/64
C		0.697	0.907	1.171	1.502	1.742	2.186
		±0.020	±0.025	±0.025	±0.025	±0.030	±0.030
D		0.960	1.140	1.445	1.770	2.060	2.545
E		1.491	1.835	2.165	2.601	2.921	3.521
F		0.135	0.150	0.170	0.200	0.220	0.250
G		1 25/32	2 3/16	2 35/64	3 3/64	3 25/64	4 3/64
H (Min.)		47/64	7/8	59/64	1 1/32	1 1/16	1 5/32
I		1.220	1.510	1.820	2.220	2.520	3.090
J		1.168	1.444	1.724	2.109	2.395	2.925
K		0.145	0.160	0.175	0.203	0.220	0.250
L (Min.)		0.145	0.160	0.180	0.210	0.230	0.260
M		0.145	0.160	0.180	0.210	0.230	0.260
O		1.345	1.650	1.975	2.395	2.710	3.310
P		0.113	0.115	0.135	0.130	0.143	0.165
Q		0.835	1.030	1.315	1.650	1.915	2.395
R		0.591	0.728	0.930	1.167	1.355	1.694
S		1.19	1.47	1.78	2.18	2.47	3.04
T		1.455	1.559	1.736	1.968	2.243	2.509
U		1.09	1.35	1.63	2.00	2.27	2.80
V		14-NPS	11-NPS	11-NPS	10-NPS	10-NPS	10-NPS
X		0.159	0.154	0.184	0.174	0.199	0.224
Y		0.831	0.904	1.049	1.219	1.369	1.604
a radius		0.094	0.125	0.125	0.125	0.125	0.125
b		0.741	0.951	1.216	1.561	1.801	2.276
c		0.972	1.174	1.494	1.828	2.116	2.638
d Pitch diameter	Max.	1.4403	1.7702	2.1001	2.5300	2.8500	3.4499
	Min.	1.4348	1.7640	2.0938	2.5233	2.8432	3.4430

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Detail dimensions of unions, inches (see figure 12, figure 13, and figure 14) – Continued.

Dimensions		NPS					
		½	¾	1	1¼	1½	2
f Pitch diameter	**Max.**	1.4420	1.7720	2.1020	2.5320	2.8520	3.4520
	Min.	1.4420	1.7720	2.1020	2.5320	2.8520	3.4520
g Major diameter	**Max.**	1.4867	1.8293	2.1592	2.5950	2.9150	3.5149
	Min.	1.4764	1.8172	2.1471	2.5821	2.9021	3.5020
h Minor diameter	**Max.**	1.4100	1.7424	2.0724	2.4992	2.8192	3.4192
	Min.	1.4111	1.7326	2.0626	2.4887	2.8087	3.4087

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

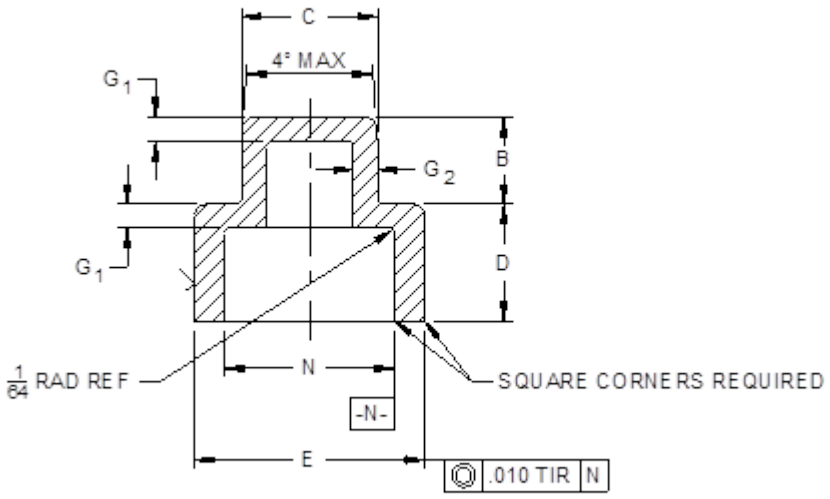


FIGURE 15. Plug-cored pattern (see table XIX).

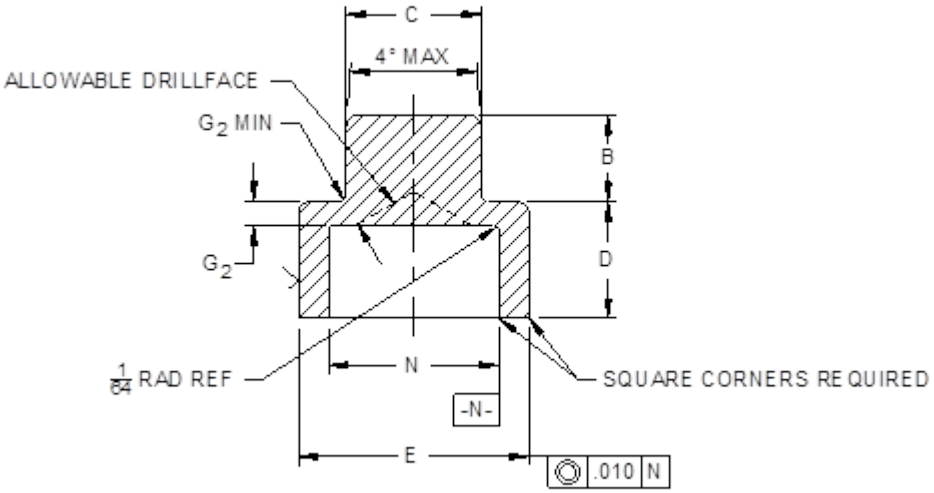


FIGURE 16. Plug-solid pattern (see table XIX).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

TABLE XIX. Detail dimensions of plugs, inches (see figure 15 and figure 16).

NPS	B	C		D		G1	G2	N	E	
	Height square min.	Across flats		Plug length		Metal thickness		Inside diameter max.	OD	
		Nominal	Max.	Min.	Max.	Top max.	Side of face max.		Min.	Max.
½	0.38	9/16	0.563	0.511	0.613	0.09	0.12	0.53	0.837	0.840
¾	0.44	5/8	0.625	0.552	0.662	0.10	0.13	0.72	1.047	1.050
1	0.50	13/16	0.813	0.594	0.713	0.11	0.14	0.93	1.312	1.315
1¼	0.56	15/16	0.938	0.666	0.799	0.12	0.15	1.25	1.657	1.660
1½	0.62	1⅛	1.125	0.801	0.961	0.13	0.16	1.47	1.897	1.900
2	0.68	1 5/16	1.313	0.842	1.010	0.15	0.17	1.91	2.372	2.375
2½	0.74	1½	1.500	0.977	1.172	0.17	0.18	2.32	2.872	2.875
3	0.80	1 11/16	1.688	1.034	1.240	0.19	0.19	2.90	3.497	3.500
3½	0.86	1⅞	1.875	1.091	1.309	0.20	0.20	3.36	3.997	4.000
4	0.92	2¼	2.250	1.142	1.370	0.22	0.22	3.83	4.497	4.500
5	0.98	2¾	2.750	1.256	1.506	0.24	0.24	4.30	5.560	5.563
6	1.05	3½	3.500	1.425	1.710	0.30	0.30	5.38	6.621	6.625

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

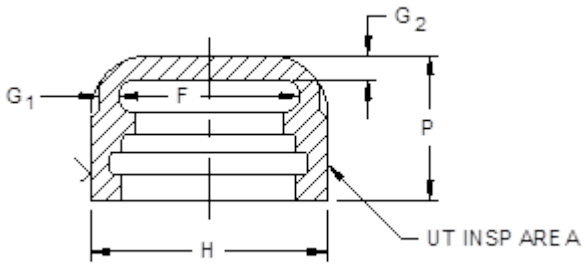


FIGURE 17. Cap.

TABLE XX. Detail dimensions of cap, inches.

NPS	F		G1	G2	P	H
	Inside diameter of fittings		Metal thickness of sides	Metal thickness of top	Height	Band diameter
	Min.	Max.	Min.	Min.		Min.
½	0.840	0.900	0.090	0.120	0.931	1.17
¾	1.050	1.110	0.100	0.130	1.011	1.42
1	1.320	1.390	0.110	0.150	1.146	1.72
1¼	1.660	1.730	0.120	0.170	1.256	2.10
1½	1.900	1.970	0.130	0.190	1.308	2.38
2	2.380	2.450	0.150	0.220	1.475	2.92
2½	2.880	2.980	0.170	0.250	1.793	3.49
3	3.500	3.600	0.190	0.290	1.956	4.20
3½	4.000	4.100	0.200	0.320	2.056	4.75
4	4.500	4.600	0.220	0.360	2.237	5.31
5	4.560	5.660	0.300	0.450	2.476	6.29
6	6.630	6.730	0.340	0.550	2.655	7.56
8	8.625	8.725	0.400	0.600	3.140	9.43
10	10.750	10.850	0.480	0.700	3.625	11.70
12	12.750	12.850	0.575	0.800	4.124	13.84

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4730-2017-092)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.